

TrES-5b

R-1388

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_141011 · created 2026-08-01T02:37:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456941.8075 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.026
Transit depth (Rp/Rs)^2	1.72 +/- 0.69 [%]
Orbital Inclination (inc)	88.16 +/- 2.26
Airmass coefficient 1 (a1)	0.999 +/- 0.022
Airmass coefficient 2 (a2)	0.0029 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	2.22 %
Best Comparison Star	#3 - [367, 259]
Optimal Method	PSF photometry
Transit Duration (day)	0.0834 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.131 ± 0.026 vs 0.142 (deviation 0.32σ)
Duration fit vs published	0.0834 d vs 0.0758 d (deviation 1.07σ)
Mid-time O–C	4.2 min
Depth SNR	3.8
Residual scatter	2.22 %

Light curve

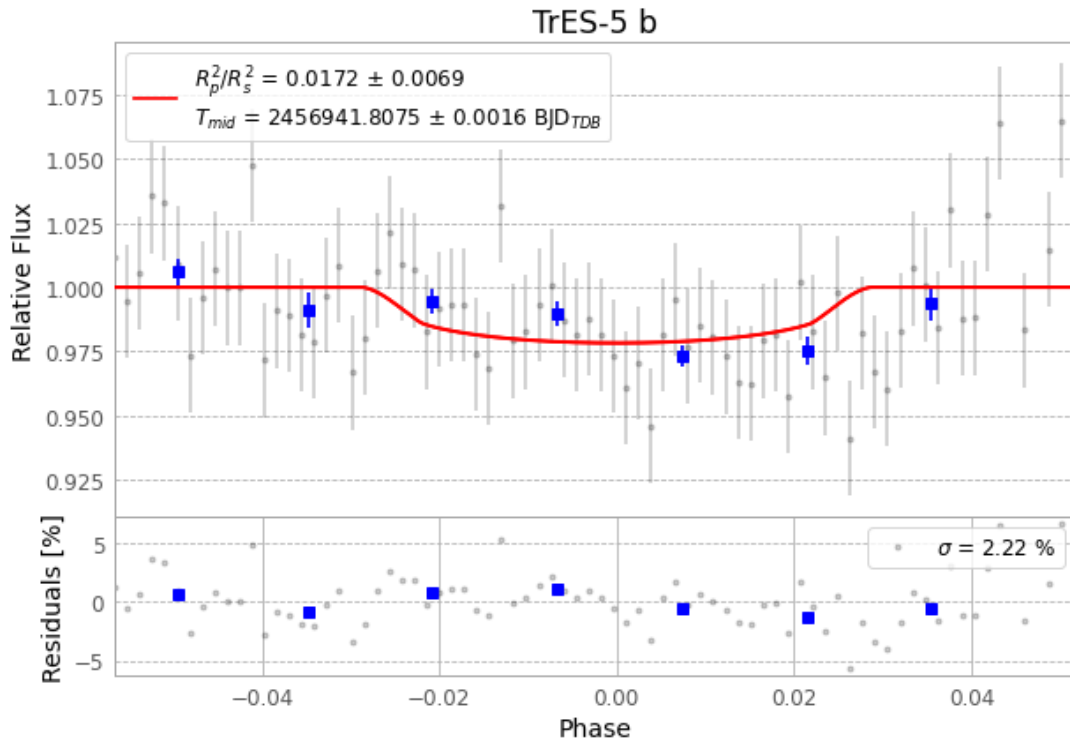


Figure 1 — FinalLightCurve_TrES-5 b_2014-10-10.png (SHA-256 d91ab1775a2f8f25...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [231, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 48a6ac95892594f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), TRES-5141011051813.FITS → TRES-5141011090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-141011.log (59bad5ec3244...), FinalLightCurve_TrES-5 b_2014-10-10.png (d91ab1775a2f...), FinalLightCurve_TrES-5 b_2014-10-10.csv (5c40d06c4fe7...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 02:37Z.